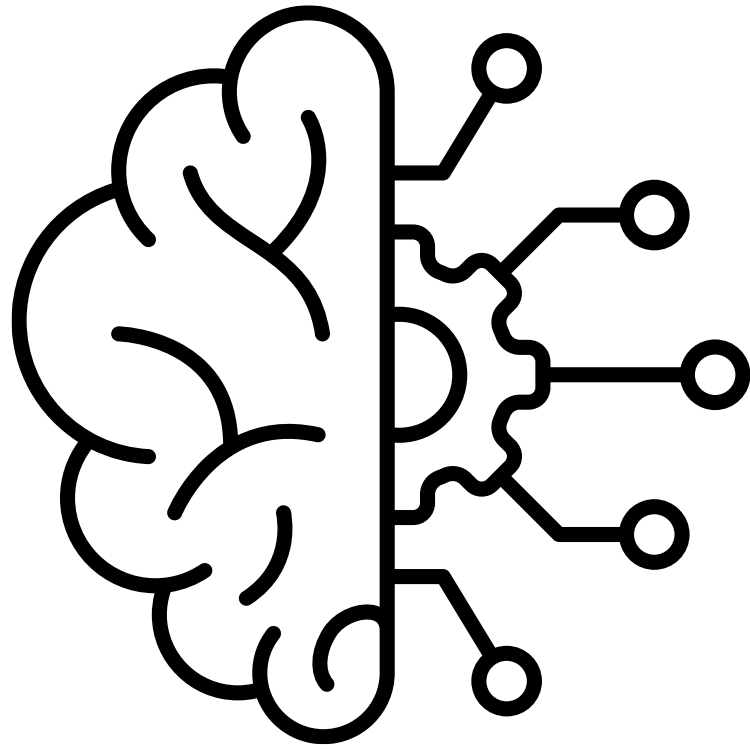




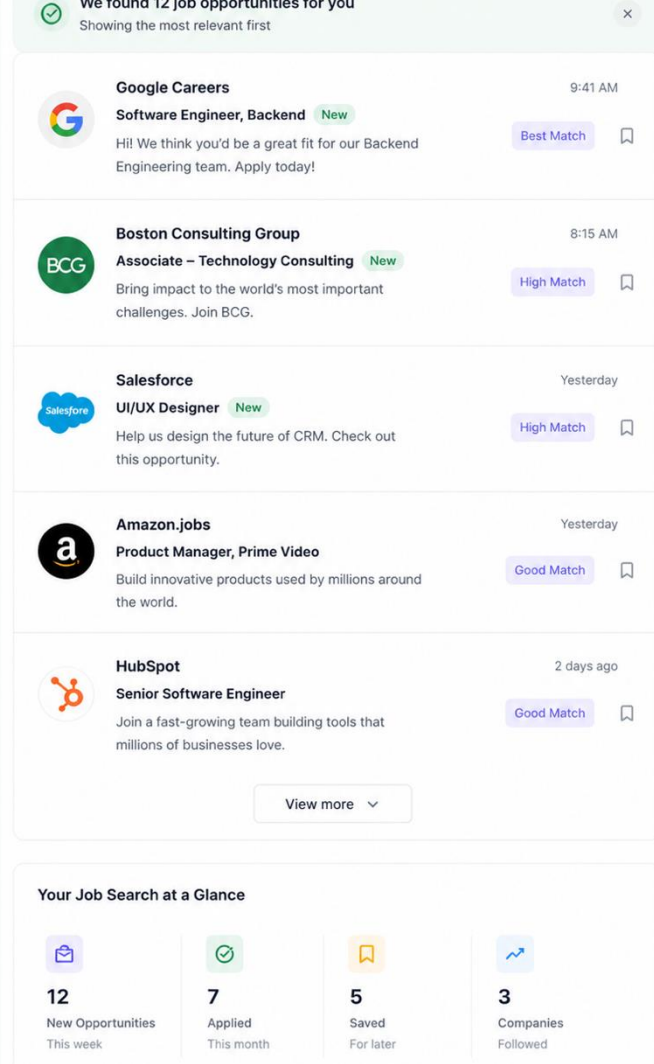
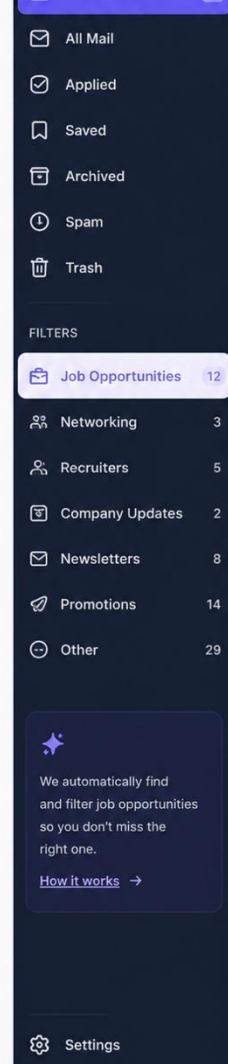
Building AI Agents for the Humanities

Session 2a: Building AI Agents

Yadira Lizama Mué, PhD

Postdoctoral Fellow, Western University
Assistant Professor of Digital Humanities

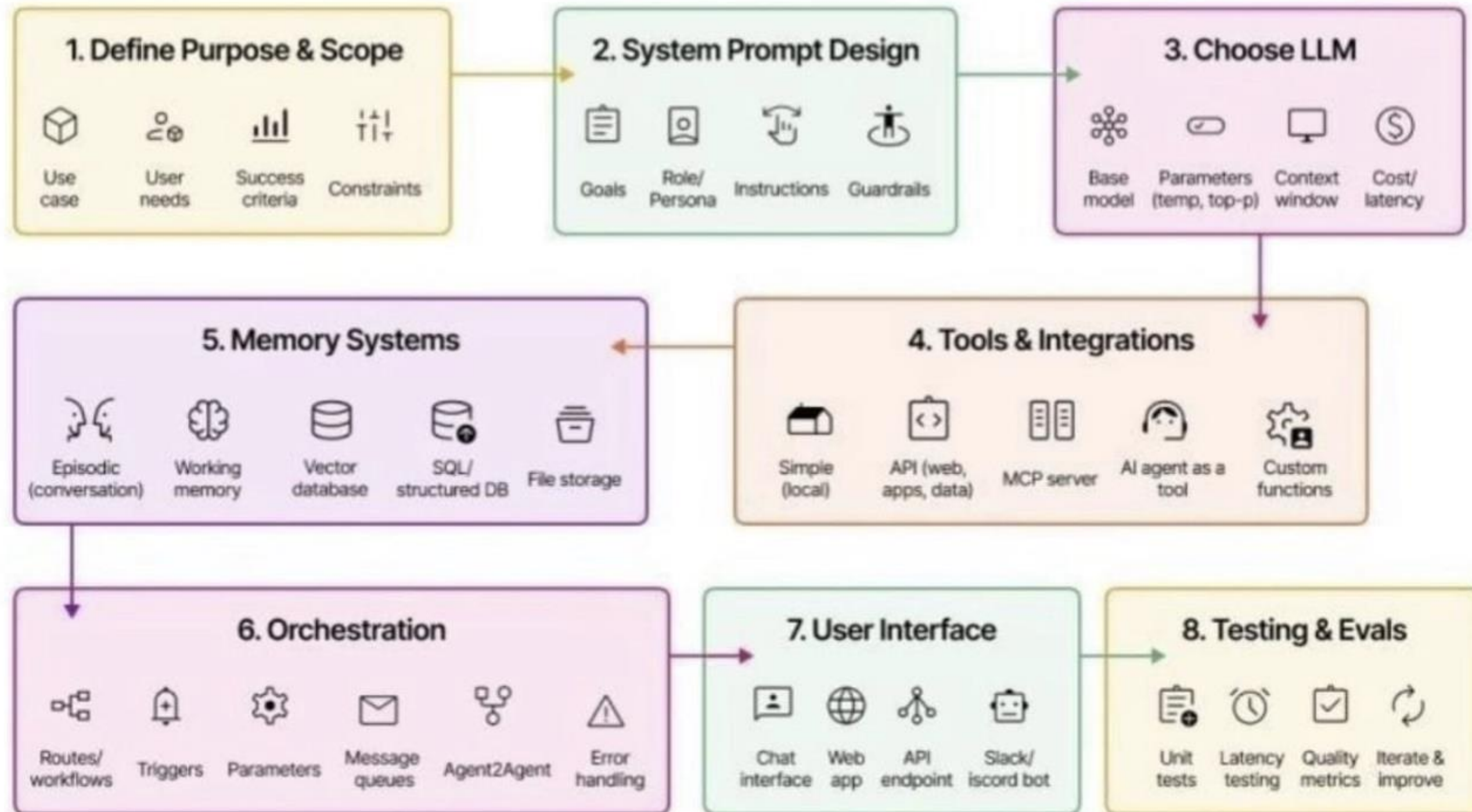




Design Challenge

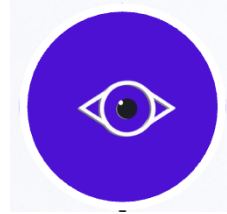
- Your agent must include:
 - **Goal:** What is the agent trying to accomplish?
 - **Prompt(s):** What instructions guide the agent?
 - **Actions:** What is the sequence of actions or decisions?
 - **Tools:** What tools or data sources does it use (e.g., search, APIs, documents)?
 - **Human Role:** Where does human input or judgment matter?

AI Agent - Components



Again... what is an AI Agent?

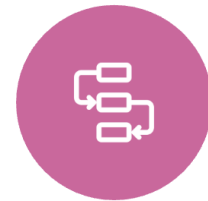
An **AI agent** is a system that:



Perceives its environment (input/data/trigger)



Decides what to do (reasoning/planning)



Acts to achieve a goal (takes actions, often over time)



Adapts based on feedback (optional but common)

**An agent perceives the
environment**

Perception - Triggers

-  **Trigger manually**
Runs the flow on clicking a button in n8n. Good for getting started quickly
-  **On app event**
Runs the flow when something happens in an app like Telegram, Notion or Airtable →
-  **On a schedule**
Runs the flow every day, hour, or custom interval
-  **On webhook call**
Runs the flow on receiving an HTTP request
-  **On form submission**
Generate webforms in n8n and pass their responses to the workflow
-  **When executed by another workflow**
Runs the flow when called by the Execute Workflow node from a different workflow
-  **On chat message**
Runs the flow when a user sends a chat message. For use with AI nodes

-  **When running evaluation**
Run a dataset through your workflow to test performance
-  **Email Trigger (IMAP) ⚡**
Triggers the workflow when a new email is received
-  **Error Trigger ⚡**
Triggers the workflow when another workflow has an error
-  **MCP Server Trigger ⚡**
Expose n8n tools as an MCP Server endpoint
-  **n8n**
Handle events and perform actions on your n8n instance →
-  **SSE Trigger ⚡**
Triggers the workflow when Server-Sent Events occur

Perception - Interaction



Human review

Request approval via services like Slack and Telegram before making tool calls



Chat



Discord



Gmail



Google Chat



Microsoft Outlook



Microsoft Teams



Send Email



Slack

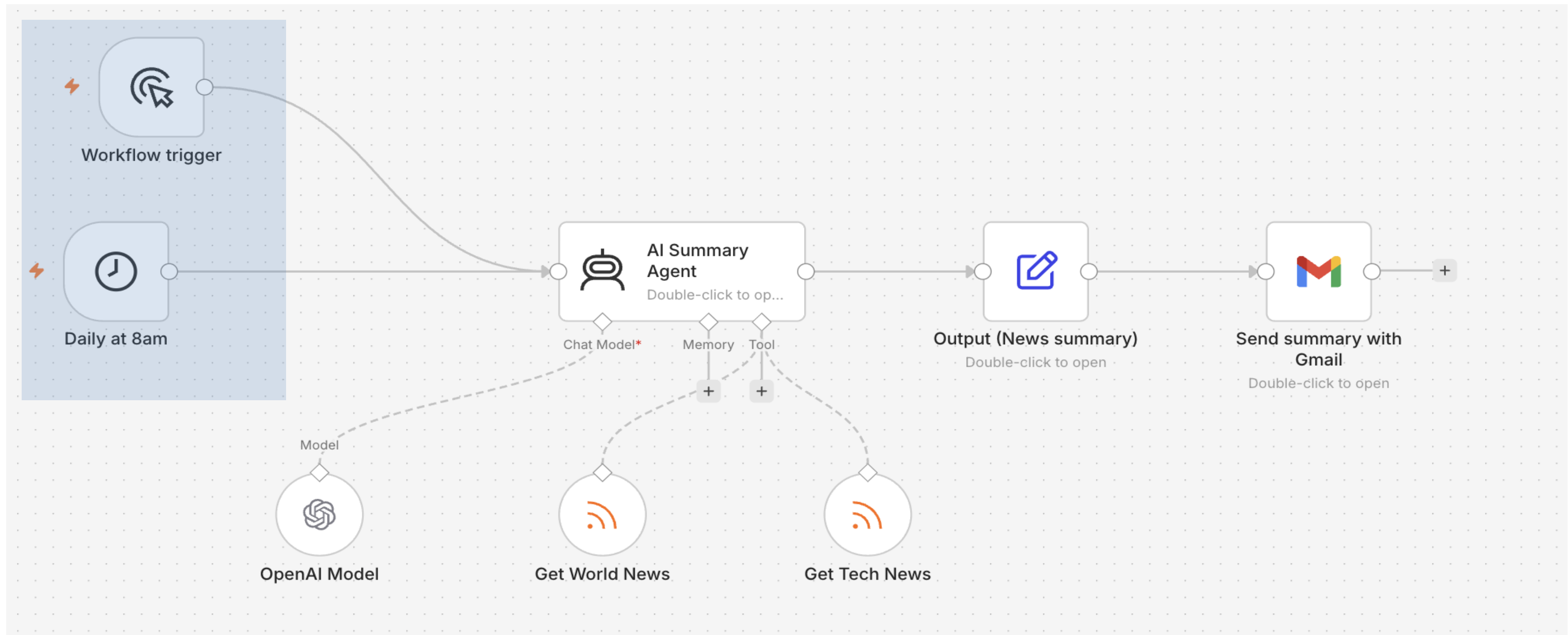


Telegram

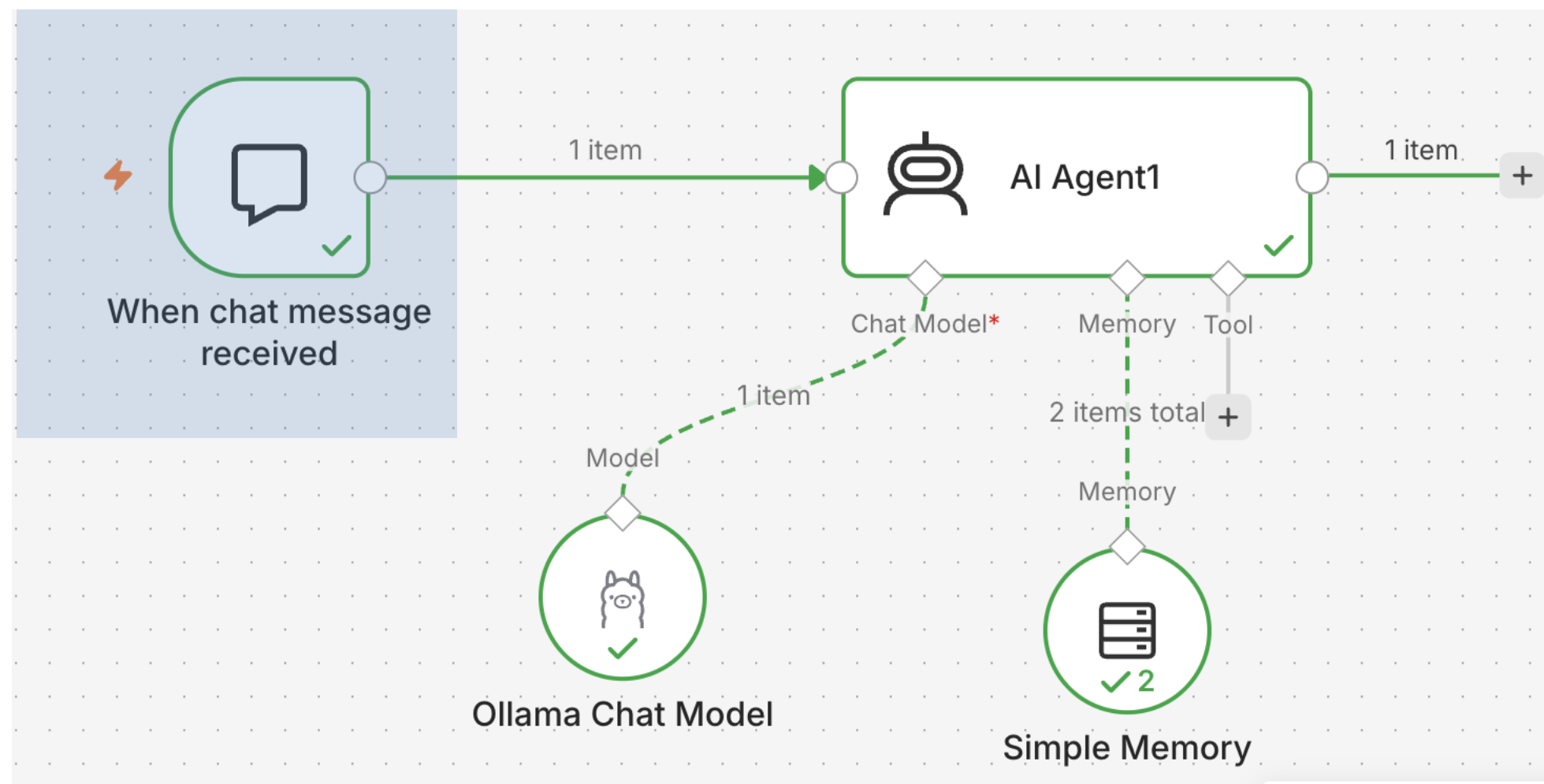


WhatsApp Business Cloud

On N8N...



On N8N...



Your turn to Practice

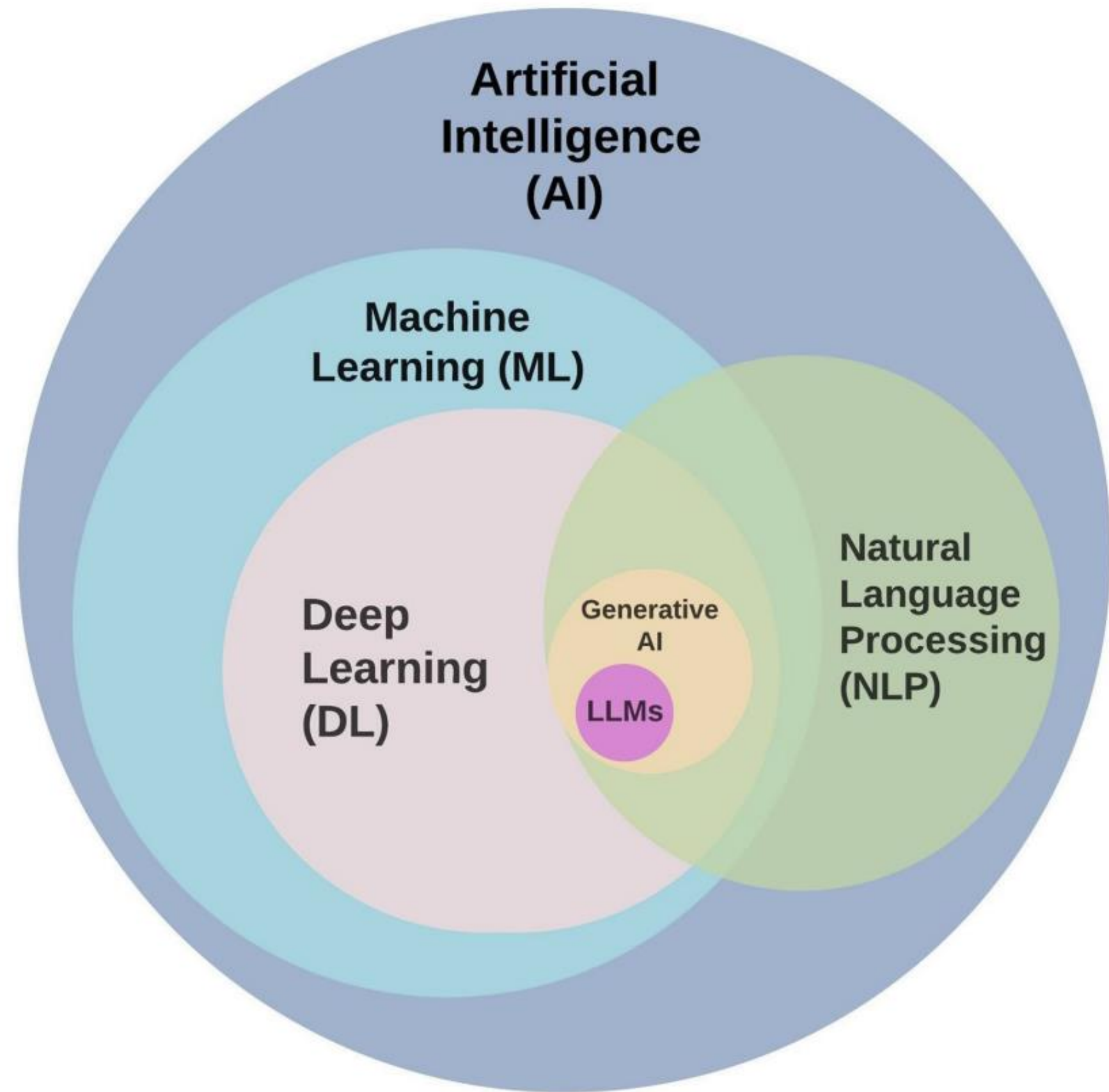
1. In groups:

- Think about how your agent perceives/communicate with the environment and write down the following:
- What triggers your agent?
- Does the agent need feedback/confirmation/approval from a human?
- Ignore the tools at this point.

2. Create a new Workflow for your project. Add and configure the trigger nodes.

**An agent must be
capable of reasoning and
planning**

Models - LLMs in the AI landscape



Cloud-Based Models

- Access to state-of-the-art capabilities
- No infrastructure burden
- Fast prototyping and iteration
- Built-in scalability
- Tool integration (function calling/actions)
- Multimodal capabilities
- Reliability and maintenance are handled for you
- Security and compliance support
- Cost efficiency (for many use cases)
- Better reasoning for agent workflows

Design intelligent behaviour (agents)
vs engineer models from scratch.

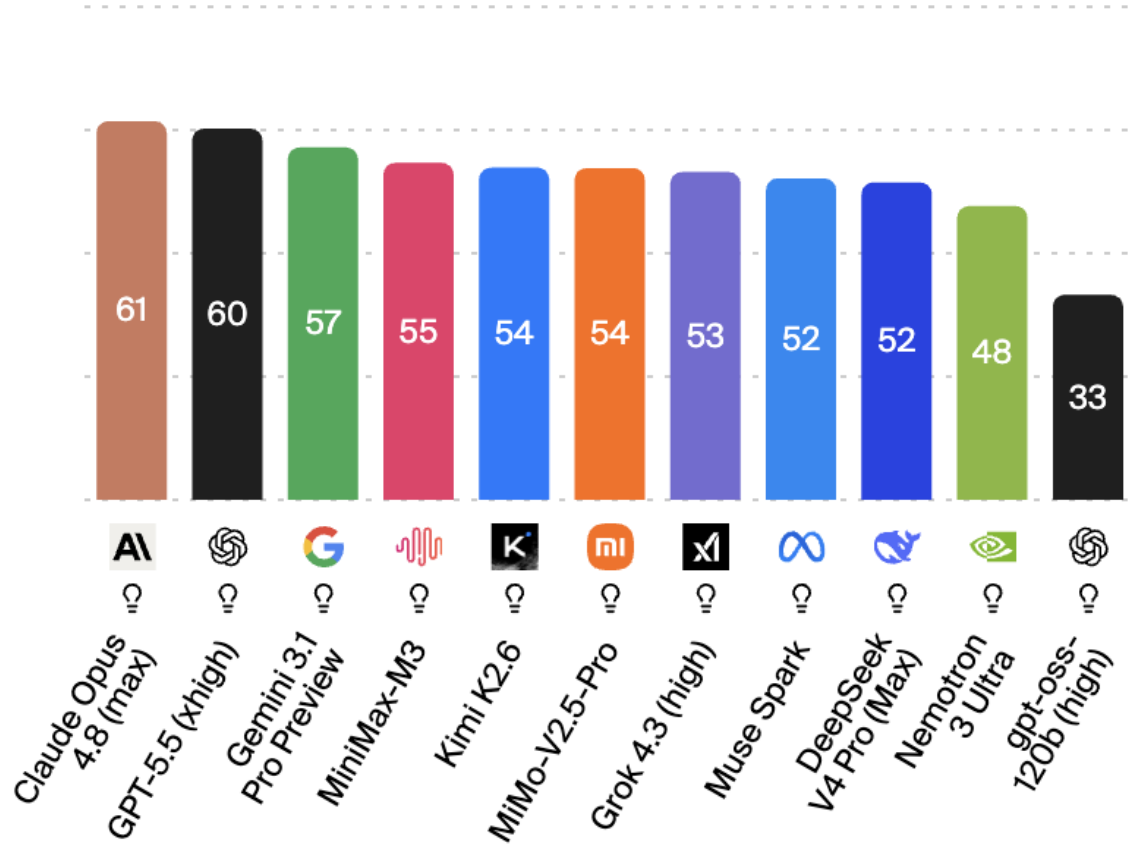
Focus on:

- prompts
- workflows
- decision logic
- tool orchestration

Cloud-Based Models

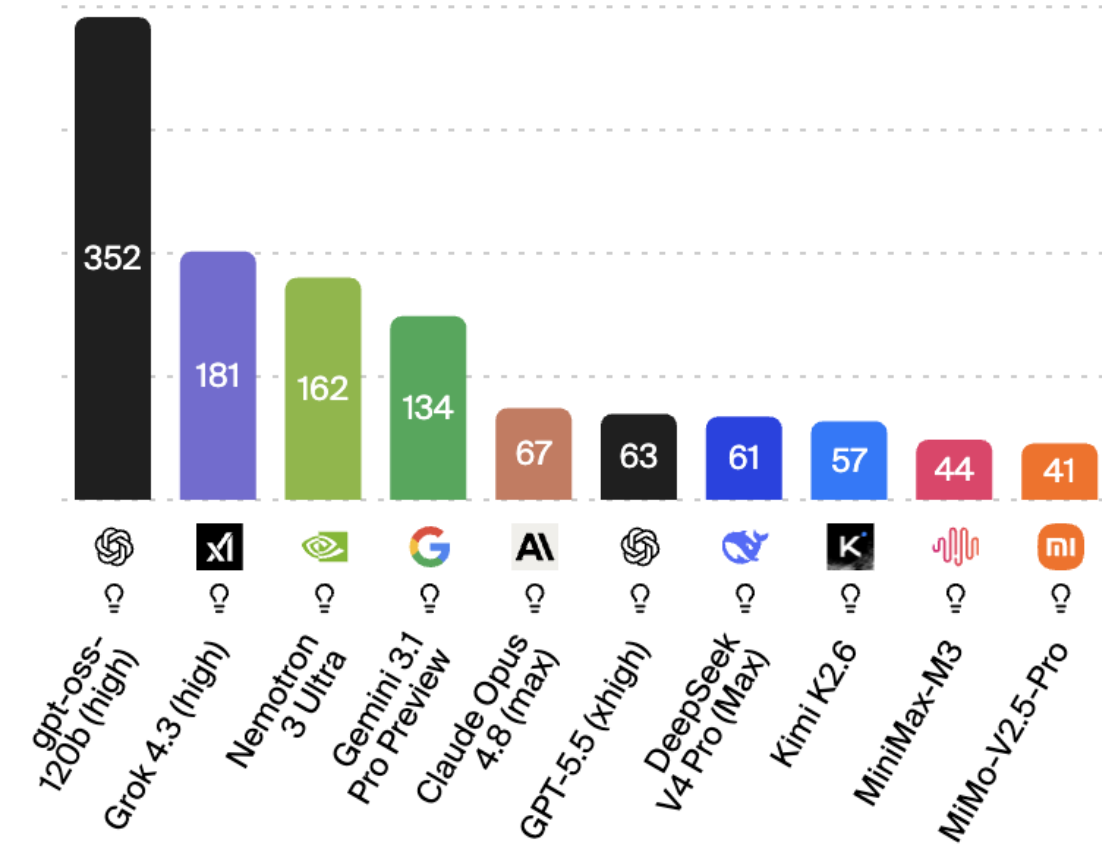
Intelligence

Artificial Analysis Intelligence Index · Higher is better



Speed

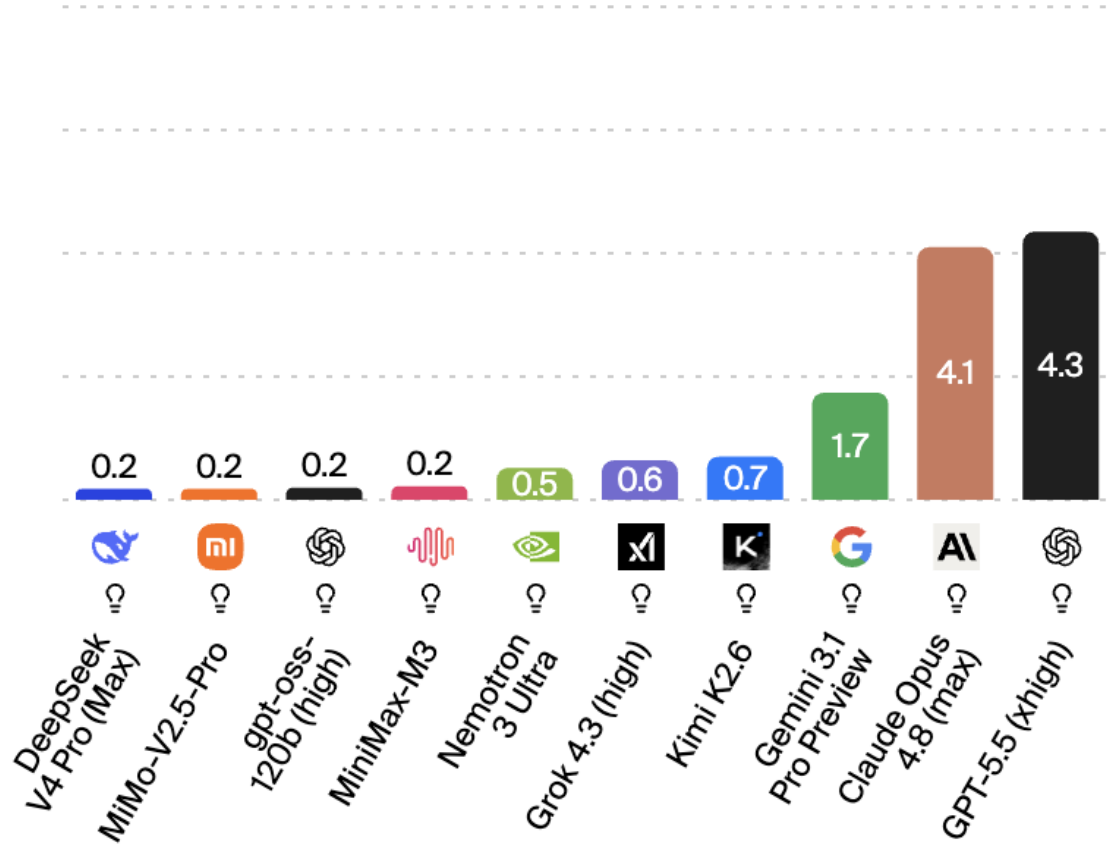
Output tokens per second · Higher is better



Price

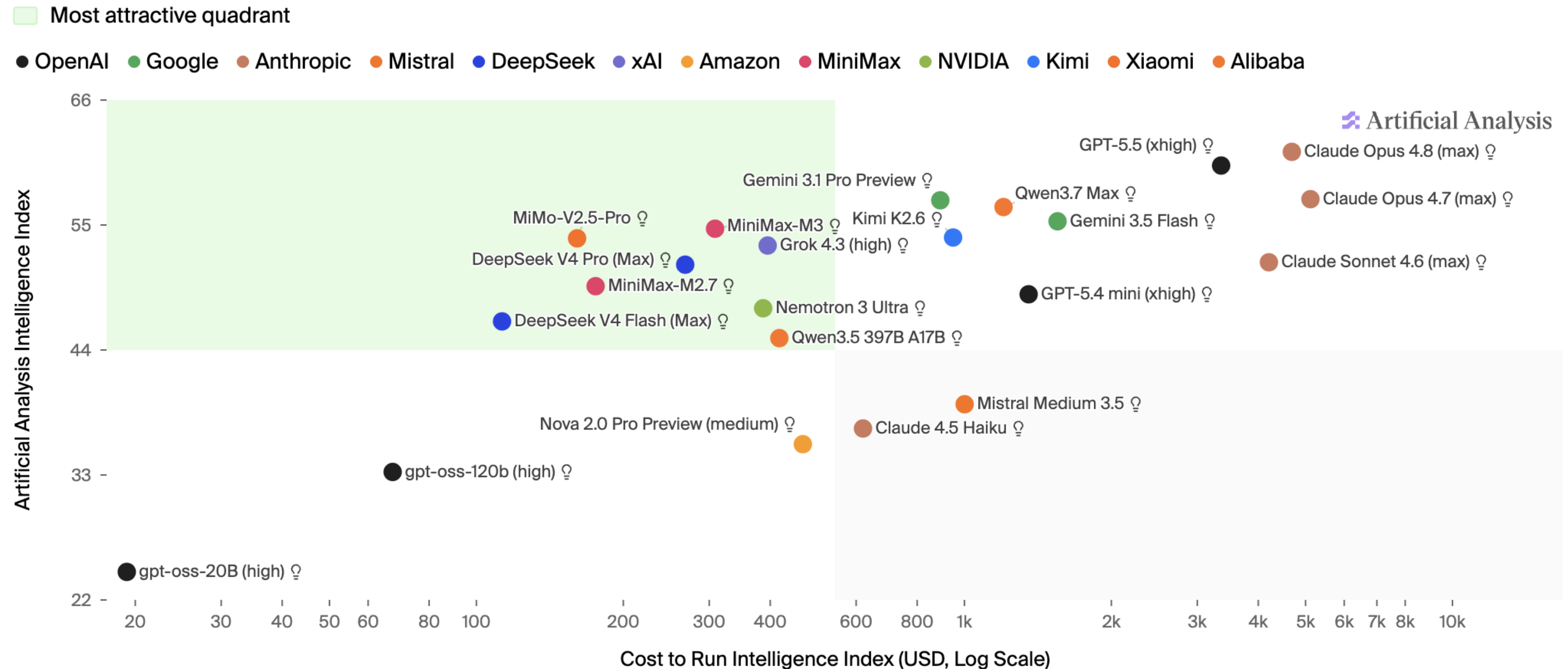
USD per 1M tokens (blended) · Lower is better

Updated



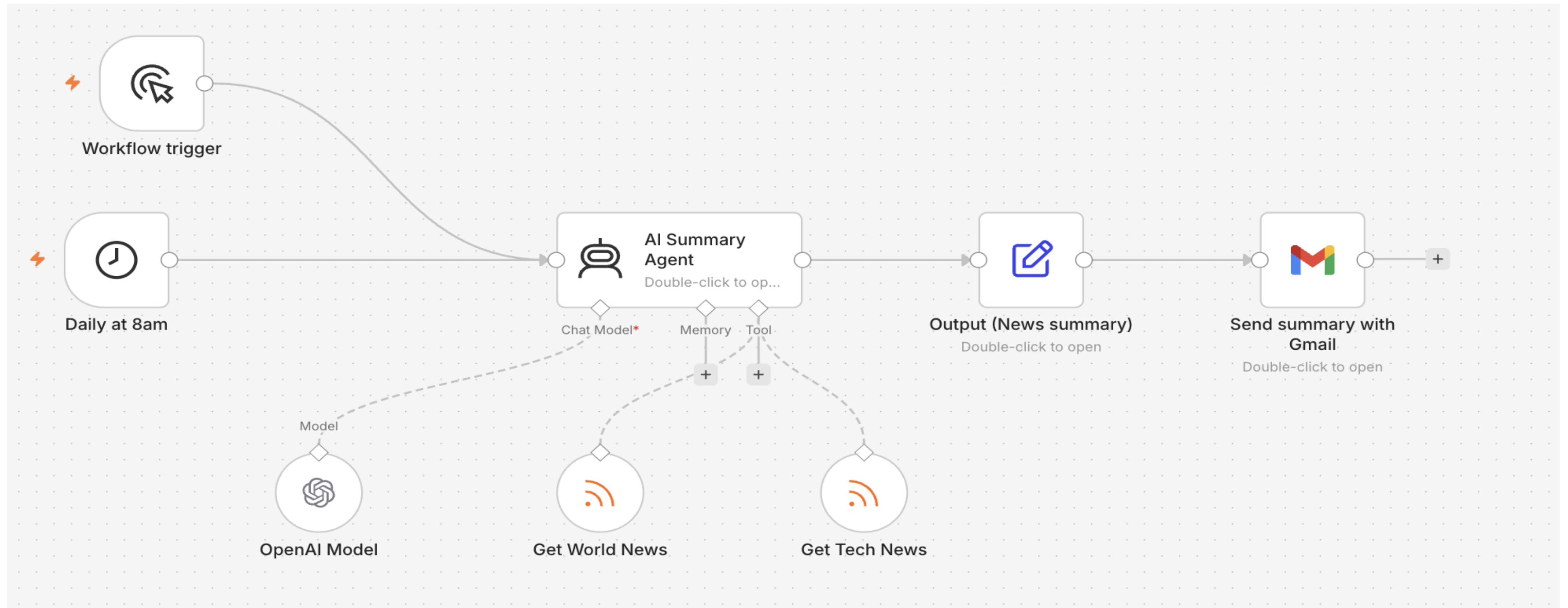
From: <https://artificialanalysis.ai/models/>

Cloud-Based Models

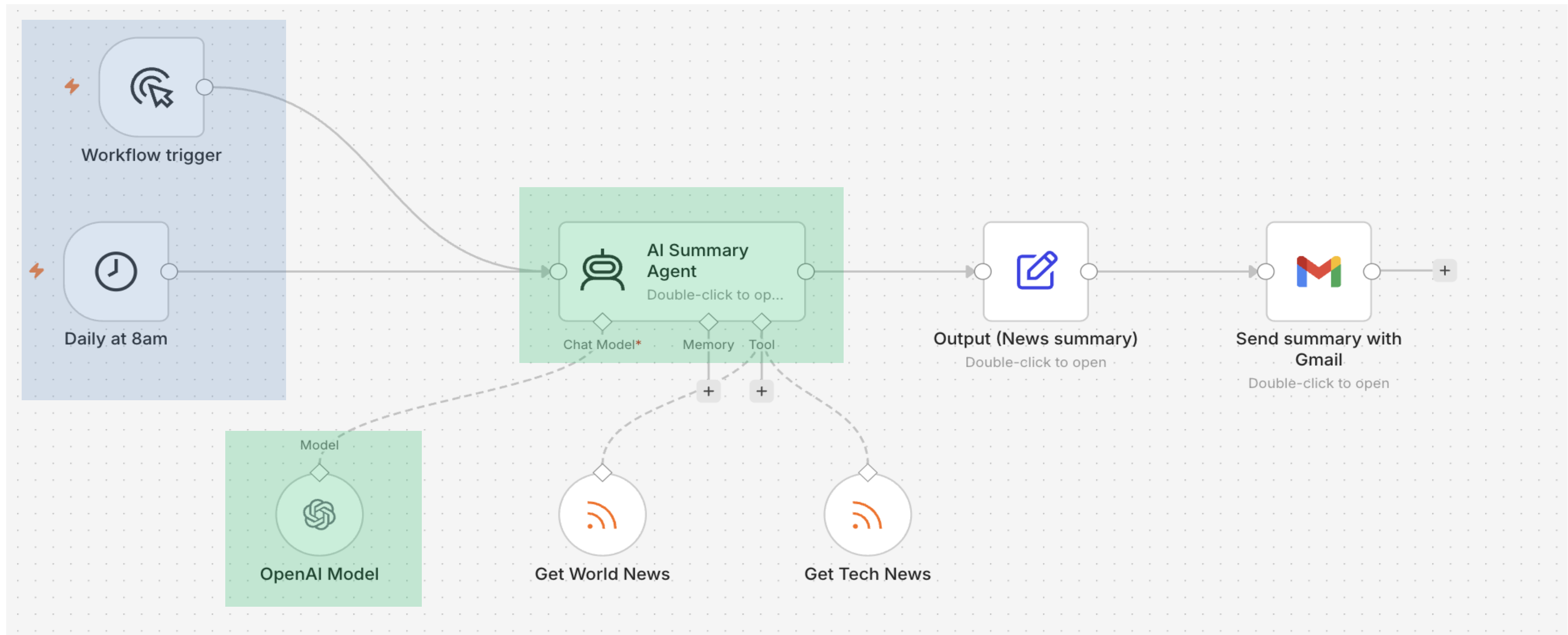


From: <https://artificialanalysis.ai/models/>

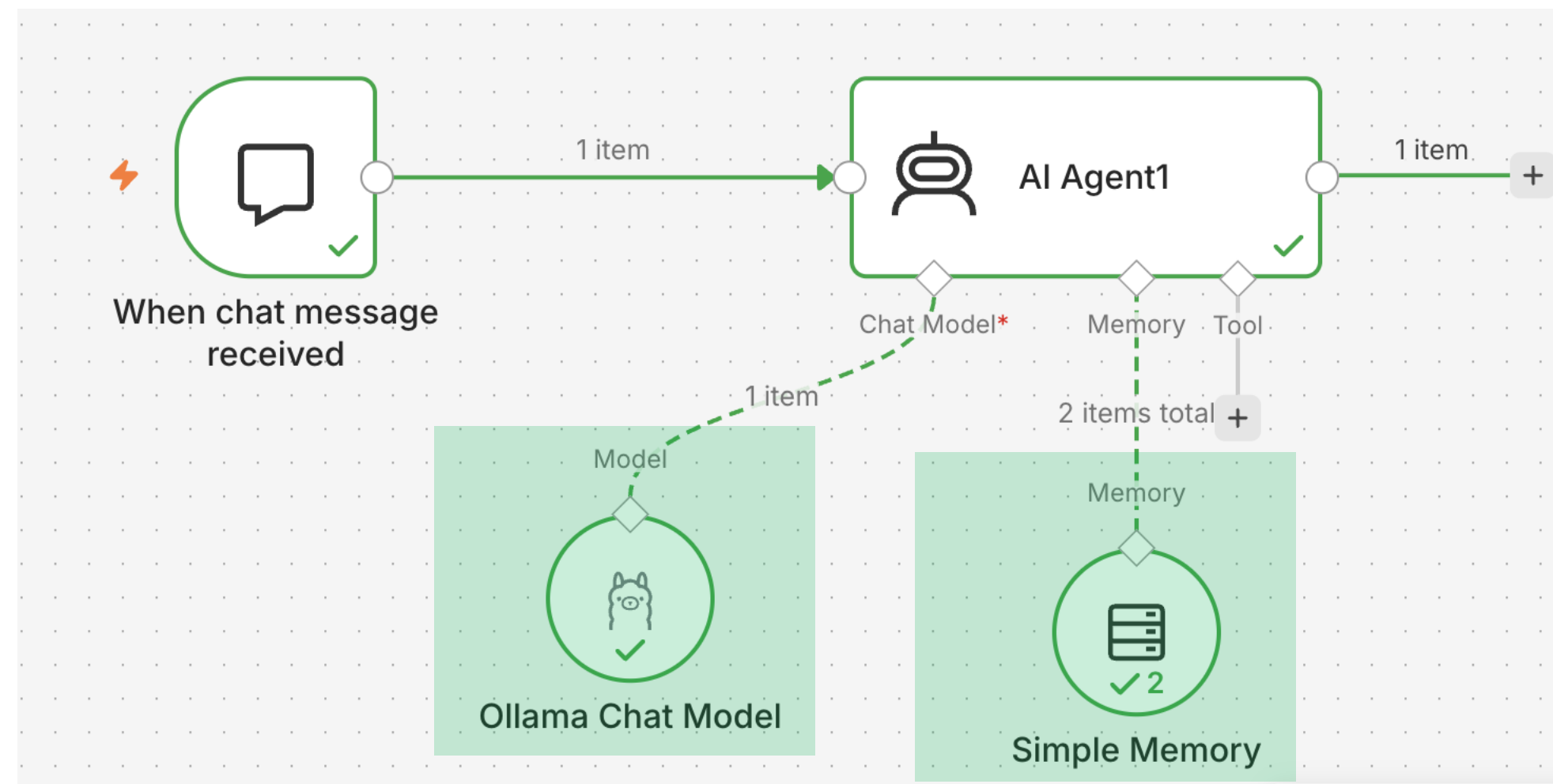
On N8N...



On N8N...



On N8N...



On N8N...

API (Application Programming Interface)

- A way for different software systems to communicate with each other
- Defines rules and endpoints for requesting and exchanging data
- Often used to access services over the internet (e.g., AI models, maps, payments)
- Let developers reuse functionality instead of building everything from scratch
- Typically works via HTTP requests (GET, POST, etc.) returning structured data (e.g., JSON)

API Key

- A unique identifier used to authenticate requests to an API
- Acts like a password/token for accessing a service
- Helps providers track usage and enforce limits or billing
- Must be kept private and secure (never exposed in public code)
- Sent with each request (e.g., in headers) to verify who is calling the API

 OpenAI account
OpenAI

Save

×

Connection

Sharing

Details

Need help filling out these fields? [Read our docs](#)

 n8n AI

 for setup instructions

API Key *

Organization ID (optional)

Only required if you belong to multiple organisations

Base URL

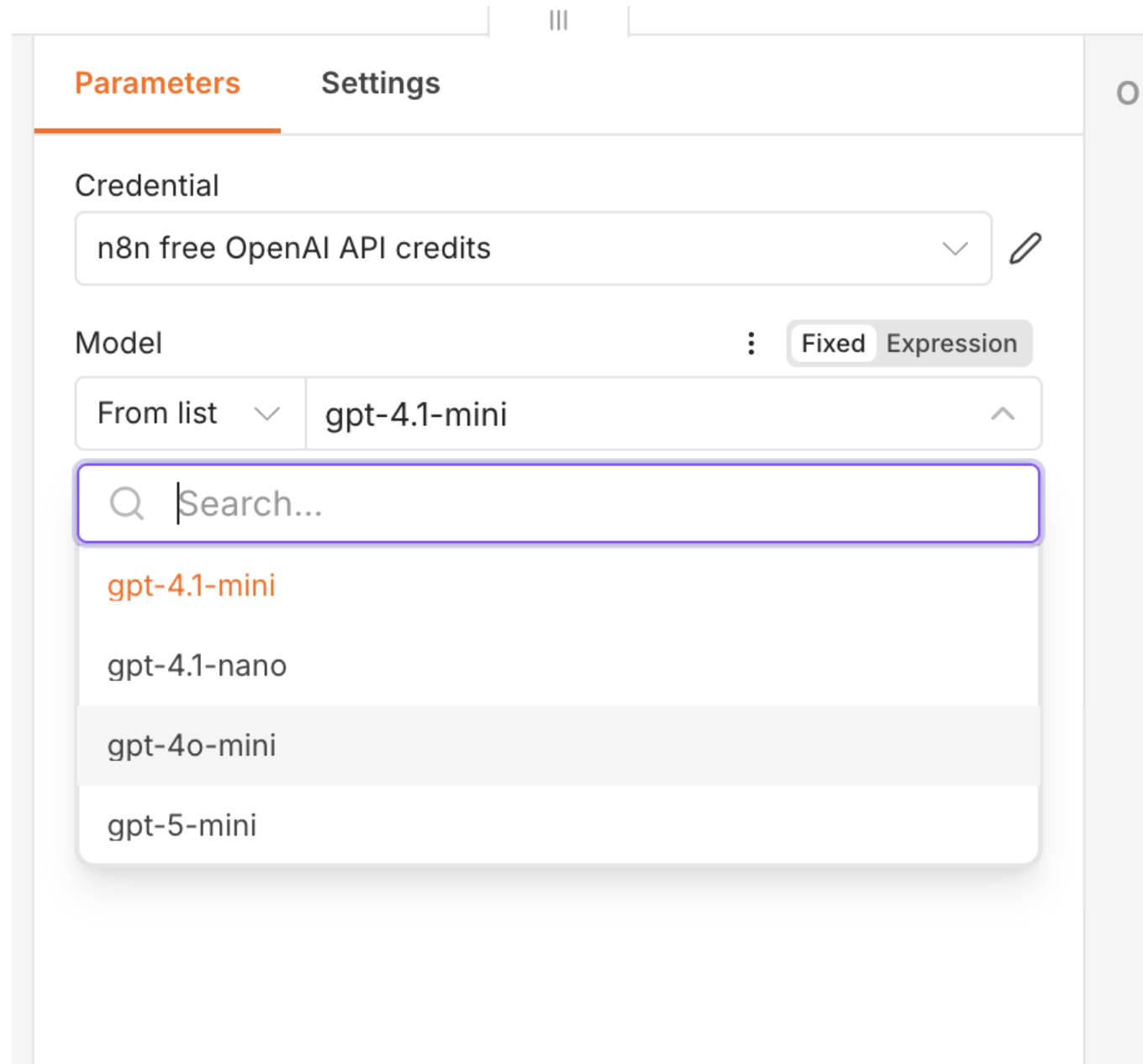
https://api.openai.com/v1

Add Custom Header

☐

On N8N...

Select one of the Models available



The screenshot shows the 'Parameters' tab of an N8N node configuration. It features a 'Credential' dropdown set to 'n8n free OpenAI API credits' and a 'Model' section with a 'Fixed' mode selector. The 'From list' dropdown is open, displaying a search bar and a list of models: 'gpt-4.1-mini' (highlighted), 'gpt-4.1-nano', 'gpt-4o-mini', and 'gpt-5-mini'.

Parameters Settings

Credential

n8n free OpenAI API credits

Model

Fixed Expression

From list

gpt-4.1-mini

Search...

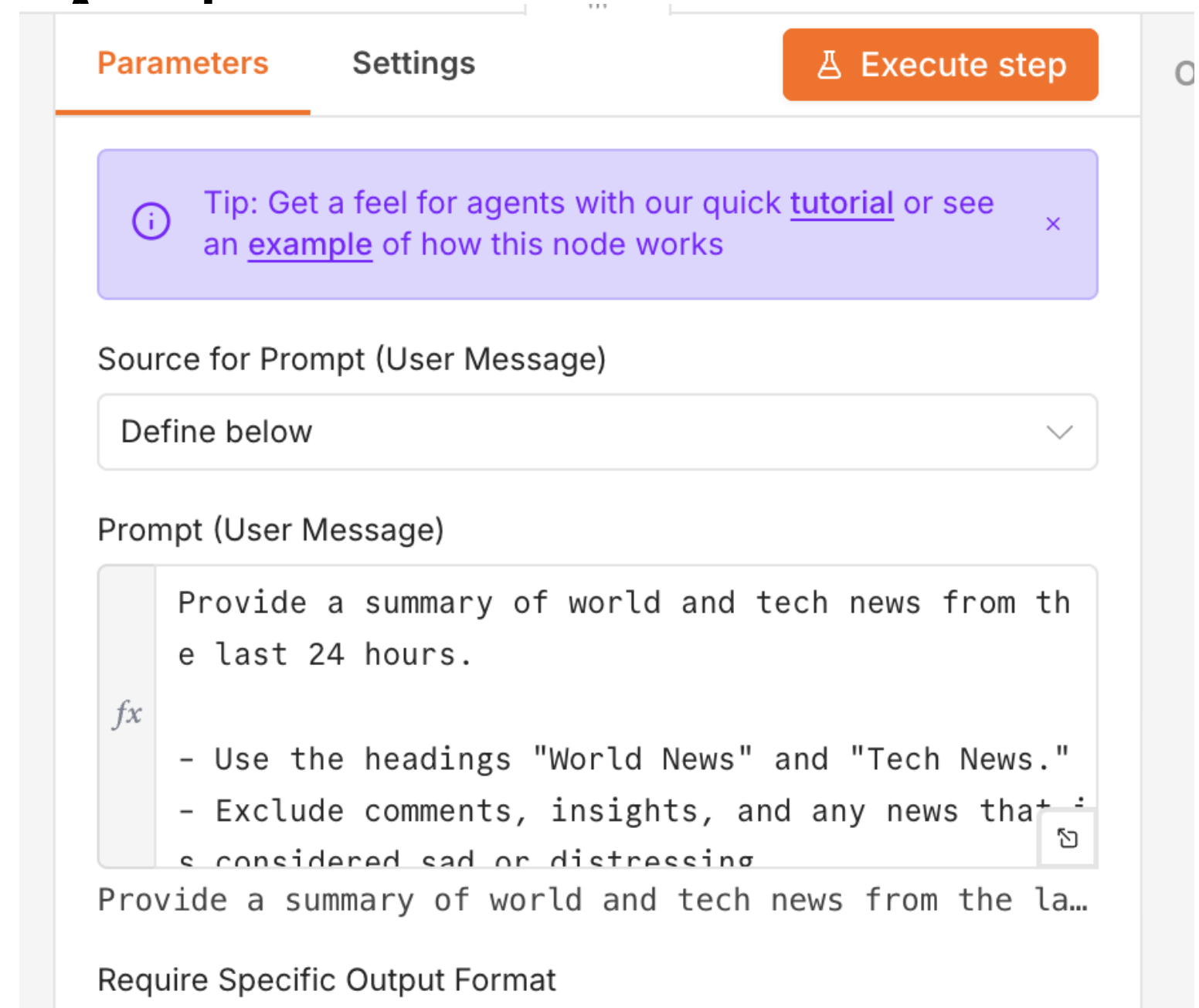
gpt-4.1-mini

gpt-4.1-nano

gpt-4o-mini

gpt-5-mini

Set Up the System Message in the



The screenshot shows the 'Parameters' tab of an N8N node configuration. It includes a tip banner, a 'Source for Prompt (User Message)' dropdown set to 'Define below', and a 'Prompt (User Message)' text area containing a system message about summarizing world and tech news. An 'Execute step' button is visible in the top right.

Parameters Settings Execute step

Tip: Get a feel for agents with our quick [tutorial](#) or see an [example](#) of how this node works

Source for Prompt (User Message)

Define below

Prompt (User Message)

Provide a summary of world and tech news from the last 24 hours.

- Use the headings "World News" and "Tech News."

- Exclude comments, insights, and any news that is considered sad or distressing.

Provide a summary of world and tech news from the la...

Require Specific Output Format

The two parts of the prompt

SYSTEM MESSAGE

The Agent's DNA

Sent with EVERY Request

Constant instruction sent with every query

Rarely or never changes

Static definition of agent identity

Contains: Role, Tools, Workflow

Fundamental agent configuration

Activates Prompt Caching

90% cost savings with n8n AI Agents

EXAMPLE

You are Support Agent.

TOOLS:

- search_docs
- create_ticket

WORKFLOW:

FAQ → search_docs
Complex → create_ticket

USER PROMPT

The specific task

Changes with every request

Dynamic data from user, webhook, or workflow

Comes from: Chat, Webhook, Node

Flexible input sources in n8n workflow

Contains ONLY specific task

No system instructions, only user content

Dynamic workflow variables

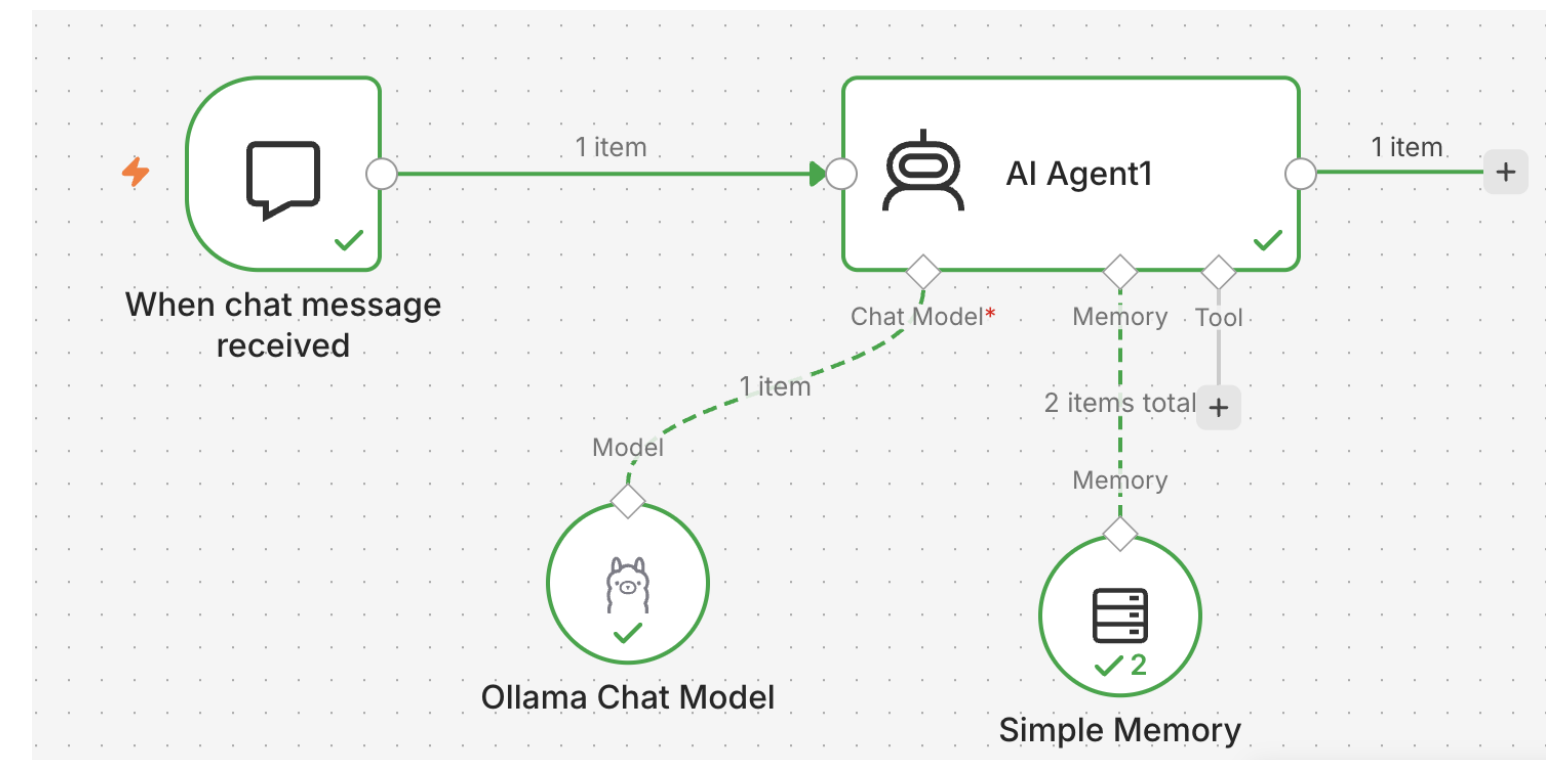
{{ \$json.message }} or other n8n Expressions

EXAMPLE

{{ \$json.message }}

// Example content:

"Where is order #12345?"



System Message (cached) → User Prompt (dynamic) → 90% Savings

Key Considerations for Agent System Message

Clarity & Specificity

"Right Altitude" Principle

Clear > Perfect

1



Classify emails intelligently and accurately



Classify emails **into**: sales, support, billing, general

URGENCY CRITERIA:

- high: contains **"urgent"**, **"asap"**, **"immediately"**, **"broken"**
- medium: time-related request without extremity
- low: everything **else**

OUTPUT: JSON

```
{  
  "category": "support",  
  "urgency": "high",  
  "confidence": 0.92  
}
```

Key Considerations for Agent System Message

Positive Instructions

DO this (not: avoid)

+57% Quality

2



Don't be too wordy.

Don't use technical jargon.

Don't make assumptions about customer intent.



Keep responses under 150 words.

Use plain language that a non-technical customer understands.

When customer intent is unclear, ask clarifying questions.

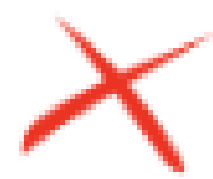
Key Considerations for Agent System Message

Few-Shot Learning

Diverse > Perfect

2-5 Examples

3



Classify the sentiment of the following messages as Positive, Negative, or Neutral.
‘I’m really impressed with the new design.’



Classify the sentiment of the following messages as Positive, Negative, or Neutral. Examples:

- ‘I love how fast this app is!’ is Positive
- ‘This update completely broke everything.’ is Negative
- ‘The app is okay, nothing special.’ is Neutral

Now classify this message: ‘I’m really impressed with the new design.’

Key Considerations for Agent System Message

Constraints & Grounding

Set clear boundaries

Anti-Hallucination

4



Answer customer support questions based **on** our documentation.



Answer customer support questions **using** ONLY information **from** the documentation

CONSTRAINTS:

- **If** information **is not in** docs: "I don't have that information in our current"
- Never make assumptions about features **or** functionality
- Never provide workarounds that aren't **documented**
- **If** multiple solutions exist: Present all documented options

ESCALATION CRITERIA:

- Customer mentions **"urgent"**, **"broken"**, **"down"** → create ticket immediately
- Question requires account-specific data → create ticket **with** details
- Documentation **is** incomplete/contradictory → create ticket noting the issue

Key Considerations for Agent System Message

Context Engineering

High-Signal Tokens

-70% Tokens

5

 You are a helpful AI assistant designed to **help** customers **with** their questions **and** concerns. You should always be polite, professional, **and** courteous **in** your responses. Make sure to read the customer's **question carefully and provide a thorough and complete answer that addresses all of their concerns. If you're not** sure about something, **it's better to say you don't** know than to provide incorrect information...



You **are** Support Agent.

RESPONSE REQUIREMENTS:

- Max **150** words
- Plain **language** (non-technical)
- Structure: Problem acknowledgment → Solution → Next steps

TOOLS:

- `search_docs(query)` → **search** product documentation
- `create_ticket(title, priority, details)` → escalate **to** human team

WORKFLOW:

1. **Search** docs **for** relevant information
2. If found: Provide answer **with** doc reference
3. If **not** found **OR** customer mentions "urgent"/"broken": **Create** ticket

Your turn to Practice

In groups, write down explicitly the System Message for the Agent that you designed yesterday in class. Consider all 5 elements:

Clarity & Specificity

1

"Right Altitude" Principle

Clear > Perfect

Positive Instructions

2

DO this (not: avoid)

+57% Quality

Few-Shot Learning

3

Diverse > Perfect

2-5 Examples

Constraints & Grounding

4

Set clear boundaries

Anti-Hallucination

Context Engineering

5

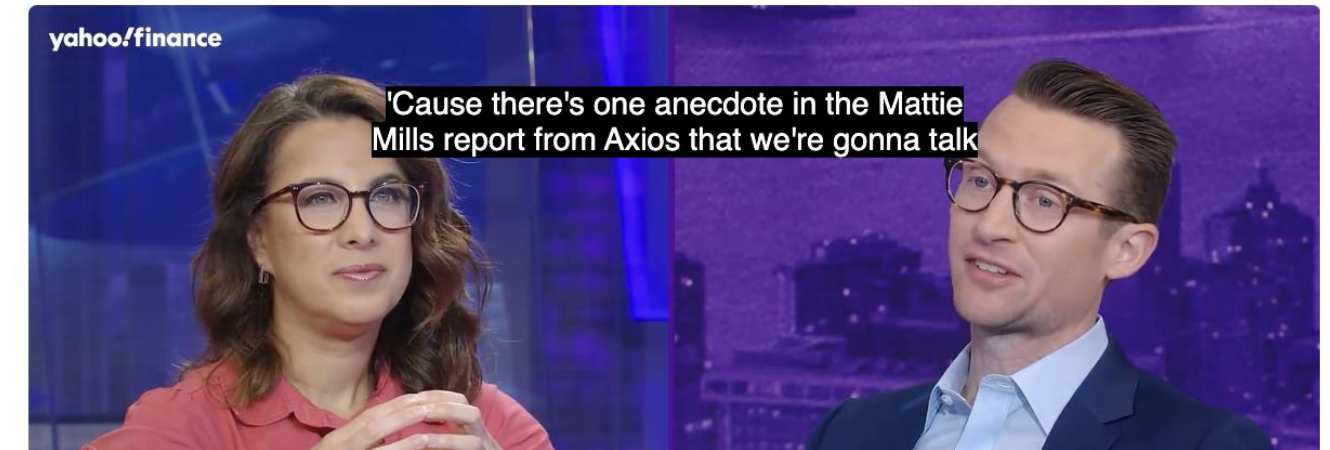
High-Signal Tokens

-70% Tokens

Disadvantages of Cloud-Based Models

- Data Privacy & Security Risks
- Dependence on Internet Connectivity
- Ongoing Costs (API Usage)
- Limited Customization & Control
- Latency & Performance Variability
- Vendor Lock-In
- Compliance & Regulatory Challenges
- Model Changes Without Control
- Limited Transparency
- Rate Limits & Quotas
- Data sovereignty risks

Microsoft cut its engineers off from AI because the bill got too big — why AI might not take your job after all



LEGAL NEWS

OpenAI Copyright Lawsuit Explained: Why Authors and Media Companies Are Suing AI Firms

By kevinlaw • April 13, 2026

OpenAI Copyright Lawsuit Explained

Written by the CredibleLaw Legal Research Team | Reviewed for legal accuracy | Updated April 2026

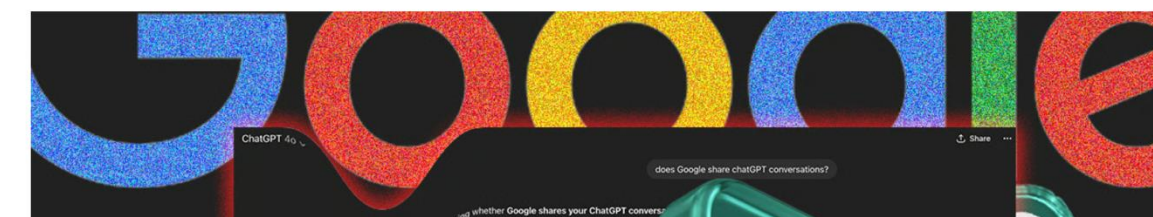
A sprawling legal confrontation between the creators of copyrighted content and the architects of artificial intelligence is reshaping the boundaries of intellectual property law in the United States. At the center of the dispute is OpenAI, the company behind ChatGPT, which faces a consolidated

Private ChatGPT conversations show up on Google, leaving internet users shocked

Published: 31 July 2025 · Last updated: 1 August 2025



Paulina Okunytė, Senior Journalist



Partner content

